

The relation between past exposure to fine particulate air pollution and prevalent anxiety: observational cohort study

SCIFACT-EVIDENCE-791050

Evidence abstract

OBJECTIVE To determine whether higher past exposure to particulate air pollution is associated with prevalent high symptoms of anxiety.

DESIGN Observational cohort study.

SETTING Nurses' Health Study.

PARTICIPANTS 71,271 women enrolled in the Nurses' Health Study residing throughout the contiguous United States who had valid estimates on exposure to particulate matter for at least one exposure period of interest and data on anxiety symptoms.

MAIN OUTCOME MEASURES Meaningfully high symptoms of anxiety, defined as a score of 6 points or greater on the phobic anxiety subscale of the Crown-Crisp index, administered in 2004.

RESULTS The 71,271 eligible women were aged between 57 and 85 years (mean 70 years) at the time of assessment of anxiety symptoms, with a prevalence of high anxiety symptoms of 15%.

Exposure to particulate matter was characterized using estimated average exposure to particulate matter <2.5 μm in diameter (PM_{2.5}) and 2.5 to 10 μm in diameter (PM_{2.5-10}) in the one month, three months, six months, one year, and 15 years prior to assessment of anxiety symptoms, and residential distance to the nearest major road two years prior to assessment.

Significantly increased odds of high anxiety symptoms were observed with higher exposure to PM_{2.5} for multiple averaging periods (for example, odds ratio per 10 $\mu\text{g}/\text{m}^3$ increase in prior one month average PM_{2.5}: 1.12, 95% confidence interval 1.06 to 1.19; in prior 12 month average PM_{2.5}: 1.15, 1.06 to 1.26).

Models including multiple exposure windows suggested short term averaging periods were more relevant than long term averaging periods.

There was no association between anxiety and exposure to PM_{2.5-10}.

Residential proximity to major roads was not related to anxiety symptoms in a dose dependent manner.

CONCLUSIONS Exposure to fine particulate matter (PM_{2.5}) was associated with high symptoms of anxiety, with more recent exposures potentially more relevant than more distant exposures.

Research evaluating whether reductions in exposure to ambient PM_{2.5} would reduce the population level burden of clinically relevant symptoms of anxiety is warranted.

Document metadata

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Evidence checklist

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points or greater on the phobic anxiety subscale of the Crown-Crisp index, administered in 2004.

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13. Research evaluating whether reductions in exposure to ambient PM_{2.5} would reduce the population level burden of clinically relevant symptoms of anxiety is warranted.

Source continuity

The canonical evidence above remains the authoritative retrieval target.